COLLABORATIVE TRANSFORMER CHATBOT PROJECT

Official Assignment Standards & Implementation Requirements

Project requirement	Standard
Participating groups	130 groups
Books per group	At least 15 books
Minimum book assignments	130 × 15 = 1,950 book assignments
Subject coverage	Different subjects across engineering, computing, science, and other approved disciplines
Primary deliverable	A validated, book-grounded chatbot from each group
Overall deliverable	One integrated multi-subject chatbot plus a documented Transformer-merging study
Computing assumption	Individual student laptops; hosted services may be used only when approved and documented

This document defines the minimum standards that every group must follow so the individual systems can be evaluated consistently and integrated into a shared platform.

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1. Project purpose and objectives

The project will involve 130 groups. Each group must build and validate a chatbot that answers questions using at least 15 books in its assigned subject area. The wider goal is to integrate the approved group systems into one multi-subject chatbot and investigate whether compatible Transformer components trained by different groups can be merged effectively.

Objective A — Build a book-grounded chatbot

- Process at least 15 books per group and preserve source metadata, including title, author, edition, and page number where available.
- Use a retrieval-augmented generation (RAG) design so the chatbot retrieves relevant book passages before producing an answer.
- Display traceable citations and avoid claiming that a source supports an answer unless the cited passage has been checked.
- State when the provided books do not contain sufficient evidence.

Objective B — Train and validate

- Build a working baseline using a shared pretrained embedding model and a language model for answer generation.
- Create a held-out evaluation set and document retrieval quality, answer correctness, citation correctness, unsupported-answer behavior, and performance.
- Where computing resources allow, fine-tune a compatible Transformer model or LoRA adapter as an additional experiment. Fine-tuning is not required merely to make the system retrieve books.

Objective C — Integrate and research merging

- Submit standardized code, document metadata, searchable passage data or an approved index, tests, evaluation results, and model configuration.
- Contribute to a unified multi-subject chatbot.
- Evaluate compatible model/adapters for merging as a separate research track. A merged model is not guaranteed to outperform separate models or a routed system.

2. Minimum scale and book-selection standards

Every group must use at least 15 books. Across 130 groups, this is a minimum of **1,950 book assignments**. This count does not imply 1,950 unique titles; duplicated titles across groups are allowed unless the coordinator publishes a separate uniqueness rule.

- Books must be relevant to the group's assigned subject and suitable for technical question answering.
- Record full bibliographic details: title, author/editor, edition, publisher, publication year, ISBN or stable identifier if available, and source/URL where legally appropriate.
- Document whether each PDF is text-based or scanned. Use OCR for scanned books when needed and inspect extracted equations, tables, figures, and page references.
- Use books only where the group has permission to process them. Do not publicly distribute copyrighted PDFs, extracted full text, or indexes unless the relevant license permits it.
- The group must keep a manifest of all 15 or more books. Books that fail extraction or are unusable must be replaced or explicitly approved as exceptions.

Suggested book manifest fields

Field	Requirement
book_id	Unique group-level identifier, e.g. G014_BOOK_003
title / author / edition	Required bibliographic fields
publisher / year / ISBN	Record when available
subject	Assigned discipline or sub-discipline
source_type	PDF text, scanned PDF, or other approved format
permission_status	Basis for permitted processing and sharing
processing_status	Pending, processed, needs OCR, failed, or approved exception

3. Mandatory system architecture

All groups must implement a consistent book-to-answer pipeline. Internal code may differ, but the outputs and interfaces must follow the standards in this document.

Stage	Minimum requirement
1. Ingestion	Read the approved book files; extract text and retain book/page metadata.
2. Cleaning and chunking	Split text into coherent passages; avoid losing headings, equations, and source boundaries.
3. Embeddings	Use the project-approved embedding model and pinned revision for comparable retrieval. If a group tests another model, keep it as a clearly labeled experiment.
4. Indexing	Build a local vector index or other approved retriever. Record the embedding model and index configuration.
5. Retrieval	Retrieve relevant passages for each question and return source metadata with the passages.
6. Generation	Generate an answer using retrieved evidence; instruct the model not to invent citations or unsupported facts.
7. Citation and abstention	Return citations that map to real stored passages; state when evidence is insufficient or conflicting.
8. Evaluation	Run automated and human-reviewed tests and save the results in the standard report format.

Recommended shared software stack

Component	Recommended tool / standard
Programming	Python; use the coordinator's supported version and pinned dependencies.
PDF text extraction	PyMuPDF; OCR for scanned books where required.
Transformers	Hugging Face Transformers with a recorded model name and exact revision.
Embeddings	Sentence Transformers using the centrally approved model and revision.
Local vector search	FAISS or another approved retriever with a common interface.
Optional fine-tuning	PEFT/LoRA on compatible base models, subject to laptop capability.
User interface	Streamlit or the project-approved interface.
Testing / collaboration	pytest, Git, and a shared repository template.

4. Required common data and software interfaces

A. Passage record

Every indexed passage must retain at least: group_id, subject, book_id, title, page (when available), chunk_id, text, and the source edition. The source metadata must travel with the text through retrieval and answer generation.

Illustrative JSON record (example values only)

```
{
"group_id": "G014",
"subject": "Electrical Engineering",
"book_id": "G014_BOOK_003",
"title": "Example Engineering Textbook",
"edition": "2nd edition",
"page": 143,
"chunk_id": "G014_BOOK_003_P143_C02",
"text": "Extracted passage text goes here."
}
```

B. Common retrieval function

```
def retrieve(question: str, top_k: int = 5) -> list:
"""Return passages with text, score, book_id, title, page, and chunk_id."""
...
```

C. Common answer response

```
{
"answer": "Answer supported by retrieved evidence.",
"sources": [
{"book_id": "G014_BOOK_003", "page": 143,
"chunk_id": "G014_BOOK_003_P143_C02"}
],
"insufficient_evidence": false
}
```

The group must ensure every returned source resolves to an actual passage. The sample IDs and textbook details above are illustrative and must not be submitted as real sources.

5. Training and validation standards

Every group must validate its system, whether or not it fine-tunes a generative Transformer.

Dataset separation

- Training set: examples used to fine-tune a model, if fine-tuning is performed.
- Validation set: used to select model settings, retrieval parameters, prompts, or checkpoints.
- Test set: kept separate until final evaluation; do not tune the system repeatedly against the test results.
- Avoid leakage from near-duplicate questions, passages, or examples. Document how the split was created.
- For a RAG-only system, the books are the searchable knowledge base; evaluation questions and reviewed reference answers form the test dataset.

Minimum evaluation categories

Category	What to test / report
Document processing	Number of books processed, failed pages, OCR issues, and metadata completeness.
Retrieval quality	Recall@5 or another agreed metric using reviewed relevant passages.
Answer correctness	Human or instructor review against a subject-specific rubric.
Citation correctness	Whether each cited book/page actually supports the answer.
Groundedness	Whether claims are supported by the retrieved evidence.
Abstention	Whether the chatbot admits insufficient evidence on designated questions.
Efficiency	Response time, peak RAM, storage use, model size, and any failure on target laptops.

Suggested initial acceptance targets

Measure	Initial target / rule
Book manifest	At least 15 books recorded per group.
Ingestion	All approved books processed or explicitly flagged with a documented exception.
Citation traceability	100% of displayed citations resolve to a real stored passage.
Retrieval Recall@5	Target at least 80% on the reviewed group benchmark; report actual results.
Answer correctness	Target at least 80% on the human-reviewed test set; report actual results.
Insufficient evidence	Report results on all designated no-answer questions; do not claim success without measurements.
Merging experiments	Do not adopt a merged model unless comparison shows benefit without unacceptable subject regressions.

These are proposed project targets, not guaranteed outcomes. The project coordinator should finalize the rubric and thresholds before formal testing begins.

6. Common evaluation benchmark

Each group must prepare a reviewed question set that reflects its subject and its 15 or more books. A recommended starting range is 30–50 questions per group, adjusted to available review time. If every group contributes 40 questions, the overall benchmark contains 5,200 questions before deduplication and quality review.

- Direct factual questions with a clear source passage.
- Reasoning questions that require combining information from more than one passage.

- Numerical or technical questions requiring calculations, formulas, or derivations where relevant.
- Cross-book comparison questions.
- Questions whose answer is not present in the assigned books.
- Citation-checking questions designed to detect fabricated or irrelevant sources.

Keep the final shared test benchmark held out from training and prompt/model selection. Use subject experts or instructors to review reference answers and the supporting pages.

7. Integration across 130 groups

The integration team should combine approved knowledge sources first, then conduct model-merging experiments separately. The project does not require all 130 systems to use separate generative models.

Recommended integration approach

- Require all groups to use the common passage schema and retrieval interface.
- Use subject-specific indexes at first, with a shared router that selects relevant subjects or searches across indexes.
- Use the same embedding model and revision for indexes that must share one semantic search space. Re-embed text if a different model is used.
- Validate duplicate books and passages, preserve the source edition, and do not discard metadata needed for citations.
- Use one shared answer-generation component where practical; add specialized models only when evaluation justifies them.
- Integrate continuously as groups pass acceptance tests rather than waiting for all 130 groups to finish.

Model merging: feasibility and constraints

What groups produce	Integration approach
Text chunks / vector indexes	Combine compatible passage records or indexes; no model-weight merging required.
LoRA adapters from the same base model	Potentially merge or combine selected adapters after checking model revision, target modules, rank, scaling, and compatibility.
Full fine-tuned models with the same architecture and starting checkpoint	Weight merging may be experimentally possible, but is resource-intensive and can cause regressions.
Different base models, tokenizers, or architectures	Not generally a direct weight merge. Use routing/ensembling or standardize the base model for a separate experiment.

Recommended LoRA experiment

Select one base model that the target laptops can realistically use. Freeze the exact model revision, tokenizer, adapter configuration, training procedure, and evaluation benchmark. Train a small pilot set of compatible adapters, evaluate each independently, then compare selected merges against the base model and the separate adapters. Scale to more adapters only if the pilot shows a measurable benefit.

The general concept can be expressed as W_merged = W_0 + sum(alpha_i × delta_W_i), where W_0 is the shared base and delta_W_i is a compatible learned update. This is a conceptual equation, not a universal recipe: LoRA representations and target modules must be handled correctly, and averaging updates may reduce performance.

A router that selects among independently validated adapters may outperform one merged model. Keep separate adapters available and do not replace the baseline unless the merged version improves the shared benchmark without unacceptable losses in individual subjects.

8. Individual-laptop resource requirements

Resource	Planning guidance
RAM	8 GB can support basic PDF processing and small RAG experiments; 16 GB or more is preferred.
Storage	Aim for at least 10–20 GB free for initial development; additional storage may be needed for model files, book data, indexes, and checkpoints.
CPU	A multi-core CPU is sufficient for document processing and many retrieval tasks; CPU-only model training may be slow.
GPU	Optional for RAG; helpful for fine-tuning. Feasibility depends on GPU memory, model size, and software support.
Internet	Needed to obtain approved dependencies/model files or use hosted inference; local workflows should be documented if offline.
API keys	Never commit credentials to Git. Use environment variables and approved account/security practices.

The group chatbot must have a documented minimum run configuration. If a laptop cannot run the selected generative model, the group may use an approved hosted inference service or a smaller quantized model, clearly noting any cost, privacy, connectivity, and availability implications.

9. Required submission package

Every group must submit all items below. Missing items may prevent acceptance or integration.

Item	Required contents
1. README.md	Group ID, subject, setup/run instructions, features, known limitations, and team roles.
2. Book manifest	At least 15 book entries with bibliographic metadata, permissions/status, and processing status.
3. Source code	Ingestion, chunking, retrieval, generation, citation handling, and application code.
4. Environment/config	Pinned dependencies, model IDs/revisions, chunk settings, embedding configuration, and reproducible run instructions.
5. Tests	Automated tests for ingestion, retrieval, citations, answer format, and insufficient-evidence behavior.
6. Evaluation dataset	Questions, reference answers, expected evidence/page, split labels, and review status.
7. Evaluation report	Actual metrics, test conditions, failures, error analysis, and limitations.
8. Index or build instructions	Approved index artifact or reproducible instructions to rebuild it from permitted sources.
9. Model artifact (if applicable)	Adapter/checkpoint, base-model revision, configuration, license, and training record. Avoid duplicate base-model uploads.
10. Demonstration	A short live or recorded demonstration with several questions and citations.

Suggested repository structure

```
group_project/
README.md
requirements.txt
config/
data/
book_manifest.json
evaluation_questions.json
src/
ingest.py
chunk.py
retrieve.py
generate.py
app.py
tests/
evaluation/
report.json
artifacts/
adapter_config.json
```

Do not upload restricted book PDFs, API credentials, personal data, or large model checkpoints to public repositories. Follow the project's rules for approved private storage and artifact sharing.

10. Suggested 12-week schedule

Weeks	Milestone	Deliverable
1–2	Standards and pilot	Common specification, starter repository, and pilot across 2–3 subjects.
3–4	Book ingestion	At least 15 books manifested per group; extraction and local indexes checked.
5–6	Individual chatbot	Working RAG chatbot, source citations, and standard interfaces.
7–8	Validation	Held-out questions, tests, reviewed answers, and group evaluation reports.
9–10	Integration	Approved indexes connected to the multi-subject router and common application.
11–12	Merging research and demo	Small adapter-merging study, cross-subject benchmark, final report and presentation.

This is a planning schedule. The coordinator may adjust it based on group size, book quality, hardware, teaching calendar, and the scale of model-merging experiments.

11. Risk controls and quality gates

Risk	Required control
Incompatible implementations	Use the shared starter code, schemas, and automated acceptance tests.
Poor OCR or missing page references	Flag extraction problems, inspect difficult pages, and replace unusable sources where necessary.
Hallucinated answers or citations	Ground answers in retrieved evidence; validate citation IDs and review factual correctness.
Insufficient laptop resources	Use lightweight embeddings, local retrieval, smaller/quantized models, or approved hosted generation.
Model merge degrades performance	Compare against base and individual models; retain the separate adapters and baseline.
Copyright or license violations	Record permissions and restrict sharing of books, extracted text, and derived artifacts as required.
Test-set leakage	Keep the final benchmark held out and document data splits.

12. Definition of completion

A group is considered complete only when it has:

- Processed at least 15 approved books or documented approved exceptions.
- Delivered a runnable chatbot using the common interface.
- Demonstrated that displayed citations map to actual passages.
- Submitted automated tests and a held-out evaluation report with actual results.
- Documented its environment, model revisions, limitations, and reproducible setup instructions.
- Submitted the approved artifacts required for integration.

The overall project is complete when the integrated multi-subject chatbot passes the shared benchmark, all accepted group contributions are traceable, and the model-merging report clearly states whether merging improved, degraded, or did not materially change performance.

Appendix A — Group submission checklist

- ■ Group ID and subject are recorded.
- ■ At least 15 books are listed in the manifest.
- ■ Bibliographic details and permissions/status are documented.
- ■ Extraction, OCR, and page references have been checked.
- ■ The common retrieval and answer interfaces are implemented.
- ■ Citations resolve to real passages.
- ■ Training, validation, and test data are separated where applicable.
- ■ The evaluation report contains actual results and limitations.
- ■ The application runs using the documented environment.
- ■ Any adapter/model artifact has a base-model revision and configuration.
- ■ No unauthorized book files, secrets, or credentials are included in public submissions.

Appendix B — Project-wide scale calculation

130 groups × 15 books per group = 1,950 minimum book assignments.

If each group contributes 40 evaluation questions, the project receives 5,200 candidate questions before review and deduplication. These figures are planning totals; unique titles and final accepted benchmark questions depend on the actual submissions.